

Lab: Cognition and Belief Optimization

Learning Goal: Analyze how attention, biases, and belief updating shape cognitive experience.

Part 1: Attention as Precision

Exercise: For each scenario, identify what is being attended to, what precision is being increased, and what is being suppressed.

1. A chess player concentrating on the board during a tournament in a noisy room
2. A lifeguard scanning a crowded swimming pool
3. A student scrolling social media during a lecture

For each, describe: What information has high precision? What has low precision? What are the consequences?

Write your response here

Part 2: Cognitive Biases in Action

Learning Goal: Identify cognitive biases in everyday reasoning and explain them using Active Inference.

Exercise: For each bias, give a real-life example, then explain it using terms from this module (precision, priors, prediction error, free energy).

1. **Confirmation bias:** When has seeking confirming evidence led you (or someone you know) astray?
2. **Anchoring effect:** Think of a negotiation or price comparison. How does the first number set the prior?

3. **Sunk cost fallacy:** Why do people continue with a failing plan rather than switching?
How does this relate to model commitment?

Write your response here

Part 3: The Dark Room Thought Experiment

Learning Goal: Engage critically with the Dark Room problem.

Discussion: Imagine an organism that truly only minimized surprise. Would it:

1. Eat? Why or why not?
2. Explore? Why or why not?
3. Have any preferences? Why or why not?

Then explain: How do preferences (prior expectations about desirable states) solve the Dark Room problem?

Write your response here

Part 4: Clinical Application

Learning Goal: Apply the framework to understand cognitive differences.

Case study: Alex is a 19-year-old college student who has recently been diagnosed with generalized anxiety disorder. Alex constantly expects the worst: failing exams, losing friends, getting sick. Even when things go well, Alex finds it hard to update these negative expectations.

Use the Active Inference framework to analyze:

- 1. What is happening with Alex's precision weighting?
- 2. Why is Alex unable to update beliefs based on positive evidence?
- 3. What might a treatment based on this framework look like?

Write your response here

Part 5: Reflection

In 150 words, reflect: Is the brain's tendency toward confirmation bias a flaw or an efficient strategy? Under what circumstances is it helpful, and when does it become harmful?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Attention analysis	Precision weighting in context
2	Bias identification	Cognitive biases as precision patterns
3	Philosophical reasoning	The Dark Room problem
4	Clinical analysis	Anxiety as aberrant precision
5	Critical reflection	Adaptive vs. maladaptive biases